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Oefeningenreeks 1B nr. 37.9

Bereken in $\mathbb{C}$ de vierkantswortel van het volgende complexe getal: $-1-2\sqrt{2}i$.

$-1-2\sqrt{2}i = (x+yi)^2 = x^2 - y^2 + 2xyi$

$\Leftrightarrow \begin{cases} x^2 - y^2 = -1 \\ 2xy = -2\sqrt{2} \end{cases}$

$\Leftrightarrow \begin{cases} x^2 - y^2 = -1 \\ xy = -\sqrt{2} \end{cases}$

$\Rightarrow \begin{cases} x^2 - y^2 = -1 \\ x^2y^2 = 2 \end{cases}$

$\Leftrightarrow \begin{cases} x^2 + 1 = y^2 \\ x^2(x^2+1) = 2 \end{cases}$

$\Leftrightarrow \begin{cases} x^2 + 1 = y^2 \\ x^4 + x^2 - 2 = 0 \end{cases}$

Laat $x^2 = \lambda$

$\lambda^2 + \lambda - 2 = 0$

$(\lambda + 2)(\lambda - 1) = 0$

$\lambda_1 = -2$ en $\lambda_2 = 1$

$x = 1$ of $x = -1$ en $y = \sqrt{2}$ of $y = -\sqrt{2}$

Aangezien $xy = -\sqrt{2}$, dan hebben x en y tegengesteld teken

Oplossing: $1 - \sqrt{2}i$ of $-1 + \sqrt{2}i$

References